Improving Multi-Task Aggression and Offensive Language Detection for Hindi–English Code-Mixed Tweets

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***Abstract*—Aggression and offensive-language detection in Hindi–English code-mixed Twitter remains challenging due to script switching, informal morphology, and pronounced class imbalance. This work extends the multi-task transformer learning (MTTL) baseline of Singh et al. (IndoNLP 2025), which couples a shared HingRoBERTa encoder with task-specific heads for three-way aggression classification (OAG, CAG, NAG) and binary offensive-language classification (OFF, NOT). We propose four targeted modifications: (i) replacement of the paper's fixed task-weighting scalar with learned homoscedastic uncertainty weighting; (ii) substitution of weighted cross-entropy with class-balanced focal loss; (iii) a hierarchical formulation in which the offence head's soft predictions condition the aggression head; and (iv) a structured error-analysis pipeline (per-class confusion matrices, joint cross-task error tables, top-confidence error mining, and gradient-based token saliency). We reproduce the baseline on the IndoNLP 2025 split (test mean macro-F1 = 0.71) and conduct a controlled hyperparameter study. Our final improved configuration, with focal loss disabled and the learning rate retuned to 1×10^{-5}, attains a test mean macro-F1 of 0.72 , a small absolute gain of 0.6 points. The more substantive contributions are methodological: the error-analysis pipeline (Improvement 4) detected a catastrophic majority-class collapse induced by focal loss with $\gamma = 2$ that scalar loss curves did not reveal, surfaced apparent annotation noise on the OAG/CAG boundary, and identified the covertly-aggressive class as the dominant residual error source. We discuss the limitations of single-seed evaluation and frame the negative result for focal loss as a substantive empirical finding.**

***Index Terms*—Multi-task learning, code-mixed text, aggression detection, offensive language detection, HingRoBERTa, focal loss, uncertainty weighting, hierarchical classification, error analysis, saliency.**

I. Introduction

Online aggression and offensive content disproportionately affect linguistic communities whose primary mode of communication is informal and code-mixed, yet the body of resources and benchmarks for code-mixed text remains comparatively small. Hindi–English code-mixed (Hinglish) Twitter is a prominent example, with users freely switching between Devanagari, romanised Hindi, and English within a single utterance. Detecting hostile language in this regime requires models that handle non-standard orthography, code-switching at the token level, and frequent neologisms borrowed from political and religious discourse.

The IndoNLP 2025 paper of Singh et al. [1] (henceforth, the *reference paper*) addresses this setting with a multi-task transformer learning (MTTL) architecture. Two related tasks , aggression classification with three labels (Overtly Aggressive, OAG; Covertly Aggressive, CAG; Not Aggressive, NAG) and offensive-language classification with two labels (Offensive, OFF; Not Offensive, NOT) , are trained jointly on top of a shared HingRoBERTa encoder [5]. The paper reports that shared layers transfer well across both objectives and that task-specific layers benefit offence more than aggression. The combined training objective is a fixed convex combination of the two cross-entropy losses, parameterised by a manually tuned scalar w_a.

This paper extends the reproduction completed in Assignment 2 by experimentally evaluating four candidate modifications to the MTTL design. Each modification targets a specific limitation observed during reproduction. We replace manual loss weighting with a learnable homoscedastic uncertainty term [2], substitute focal loss [3] for weighted cross-entropy, introduce a hierarchical conditioning of the aggression head on the offence head, and add a structured error-analysis and explainability pipeline.

The contributions of this paper are as follows. *First*, we provide an empirical comparison between the paper-faithful baseline and our improved configuration on identical splits. *Second*, we report a substantive negative result: focal loss with $\gamma = 2$ induces majority-class collapse on this task , a phenomenon that loss curves do not reveal but that our error-analysis pipeline detects immediately. *Third*, we show that the manual task-weighting scalar w_a can be replaced with a learnable homoscedastic uncertainty parameterisation without measurable loss in performance, removing a per-dataset tuning step. *Fourth*, we present a per-class and cross-task error decomposition that identifies the covertly-aggressive class as the dominant bottleneck and surfaces likely annotation inconsistency on the OAG/CAG boundary.

II. Background and Related Work

A. Aggression and Offence Detection in Code-Mixed Text

Early work on Hindi–English aggression detection used static word embeddings combined with classical classifiers or LSTM encoders [8]. These approaches were limited by the

out-of-vocabulary problem: a substantial fraction of romanised Hindi tokens never appear in English-only embedding tables, and the informal spellings used on Twitter (*nhi*, *kya*, *accha*) generate further sparsity. The introduction of multilingual transformer encoders such as XLM-R [4] substantially improved performance on code-mixed inputs through sub-word tokenisation, which gracefully handles unseen romanisations. HingBERT and HingRoBERTa [5] , XLM-R variants continually pre-trained on Hinglish corpora , are now standard backbones for Hindi–English code-mixed classification. The IndoNLP 2025 paper [1] adopts HingRoBERTa as its encoder and introduces the multi-task formulation that this paper builds upon. Closely related work on hate speech and abusive language in code-mixed data has consistently shown that a continually pre-trained multilingual encoder outperforms an equivalent monolingual model fine-tuned with code-mixed data, motivating our choice not to vary the backbone.

B. Multi-Task Learning for Related Classification Tasks

When two tasks share a feature representation, multi-task learning can regularise the encoder and exploit cross-task correlations. The reference paper combines aggression and offence losses through a fixed scalar

$$\mathcal{L} \;=\; w_a\,\mathcal{L}_a + (1 - w_a)\,\mathcal{L}_o, \tag{1}$$

selected by grid search ($w_a = 0.4$ in the published version). This formulation has two limitations. First, it is sensitive to the chosen scalar: w_a effectively trades aggression performance for offence performance, and the optimal trade-off depends on dataset balance, relative task difficulty, and label correlation. Second, it provides no theoretical guidance for unseen domains, requiring the user to re-tune w_a whenever the data distribution shifts. Kendall, Gal, and Cipolla [2] proposed an alternative in which each task is associated with a learnable log-variance parameter; tasks with higher noise automatically receive smaller effective weight, and the additive log-variance term prevents the trivial solution of inflating one variance to drive the loss to zero. We adopt their formulation as Improvement 1.

C. Loss Functions for Class Imbalance

The standard remedy for class imbalance is per-class loss weighting, where the per-example loss is multiplied by an α-vector derived from inverse class frequencies. Focal loss [3] extends this with a multiplicative term $(1 - p_t)^\gamma$ that down-weights well-classified examples and concentrates gradients on hard ones. The original paper recommended $\gamma = 2$ for highly imbalanced object-detection settings (millions of easy negatives), and subsequent work has explored the technique on other dense-prediction problems. Application to text classification at the scale of this dataset is less well-studied, and the published evidence is mixed: some studies report small gains on highly skewed text tasks, while others report no benefit or regressions when class imbalance is moderate. We evaluate focal loss as Improvement 2 with attention to whether the imbalance regime in our dataset matches that of the original proposal.

D. Hierarchical Multi-Label Classification

Where two labels are correlated, hierarchical decompositions can let the easier sub-task condition the harder one. In our setting almost all overtly-aggressive (OAG) tweets are also offensive (OFF), while covertly-aggressive (CAG) tweets are split between OFF and NOT (see Table I). This suggests that predicting offence first , a two-way decision , can provide a useful signal to the aggression head. A common pitfall in hierarchical decompositions is the *teacher-forcing* mismatch: training on gold offence labels but inferring with predicted offence labels produces a distribution shift that can hurt downstream performance. We avoid this by conditioning on *soft* probabilities at both train and inference time, which keeps the two computations identical and lets gradients flow through the offence head from the aggression loss.

E. Explainability for Text Classifiers

Gradient-based saliency methods [7] attribute model predictions to individual input tokens by combining the input embedding with its gradient with respect to a chosen output logit. They are inexpensive (one additional backward pass per example) and require no auxiliary model or training procedure, which makes them well-suited to post-hoc analysis of an already-trained classifier. More principled methods such as Integrated Gradients and SHAP exist and are direct drop-in replacements; we adopt the simpler gradient $\times$ input variant in Improvement 4 as part of a broader error-analysis pipeline, and note the limitation in Section VI-D.

III. Proposed Model

A. Architecture Overview

We adopt the MTTL architecture of the reference paper as the starting point and modify it along four orthogonal axes. The shared backbone is the `l3cube-pune/hing-roberta` encoder [5] , an XLM-R base model continually pre-trained on Hinglish corpora. A tokenised tweet is passed through the encoder to produce a sequence of contextualised representations $H \in \mathbb{R}^{T \times d}$ with $d = 768$. The pooled `[CLS]` representation $h_0 \in \mathbb{R}^d$ feeds two heads.

- An **offence head** first applies $K = 3$ task-specific blocks (Linear $\rightarrow$ LayerNorm $\rightarrow$ ReLU $\rightarrow$ Dropout) followed by a binary classifier.
- An **aggression head** applies a single classification head over either h_0 alone (parallel formulation) or $h_0 \oplus \sigma(\mathbf{z}_o)$, the concatenation of h_0 with the softmax-normalised offence logits (hierarchical formulation, Improvement 3).

A schematic of the proposed architecture is shown in Figure 1.

B. Improvement 1: Dynamic Task Weighting

The fixed scalar w_a in (1) requires manual tuning per dataset. Following Kendall et al. [2], we introduce two learnable log-variance parameters s_a and s_o representing the

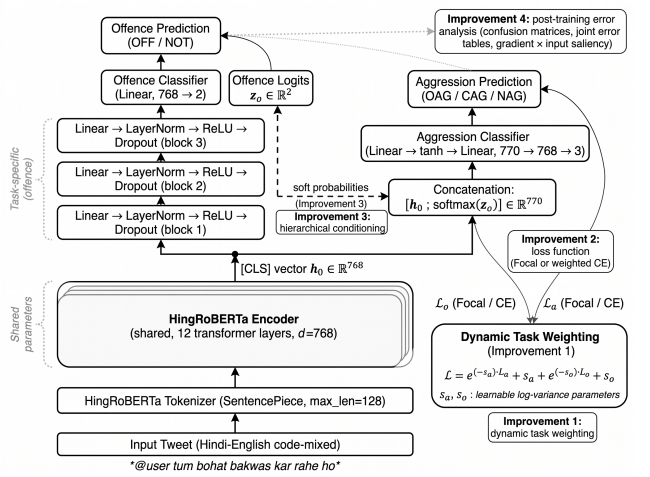

Fig. 1: Proposed MTTL architecture. The shared HingRoBERTa encoder feeds an offence head and an aggression head; in the hierarchical variant (Improvement 3) the aggression head is additionally conditioned on the offence head's soft probabilities $\sigma(\mathbf{z}_o)$. The dynamic weighting module (Improvement 1) combines the per-task losses through two learnable log-variance parameters s_a and s_o.

homoscedastic uncertainty of each task. The combined loss becomes

$$\mathcal{L} \;=\; e^{-s_a}\,\mathcal{L}_a + s_a \;+\; e^{-s_o}\,\mathcal{L}_o + s_o. \tag{2}$$

The first and third terms scale per-task losses by their precision σ^{-2}; the additive s_a and s_o terms penalise trivially small precisions and prevent the model from minimising the loss by inflating one variance. The two parameters are added to the AdamW parameter group jointly with the network weights and are initialised at zero ($\sigma_a = \sigma_o = 1$, equivalent weights).

C. Improvement 2: Class-Balanced Focal Loss

Both classification heads exhibit class imbalance. Standard weighted cross-entropy multiplies per-example losses by an α-vector derived from class frequencies. Focal loss [3] additionally down-weights confidently-classified examples:

$$\mathcal{L}_{\text{focal}}(p_t) \;=\; -\,\alpha_t\,(1-p_t)^{\gamma}\,\log p_t, \tag{3}$$

where p_t is the predicted probability of the true class, α_t is the per-class weight, and $\gamma \geq 0$ is the focusing parameter. We initialise with $\gamma = 2$ following the original paper. The α-vectors are reused from the baseline and remain frozen.

D. Improvement 3: Hierarchical Task Conditioning

In the training set, almost all OAG examples are labelled OFF, while CAG examples are split between OFF and NOT (see Section IV-A). This correlation motivates predicting offence first and conditioning aggression on its soft output. We replace the aggression head's input with

$$\mathbf{x}_a \;=\; h_0 \,\oplus\, \mathrm{softmax}(\mathbf{z}_o) \;\in\; \mathbb{R}^{d+2}, \tag{4}$$

where $\mathbf{z}_o$ are the offence logits. Soft probabilities (rather than hard arg-max labels) are used at both training and inference time so that the train-inference computation is identical, and so that gradients can flow through the offence head from the aggression loss.

E. Improvement 4: Error Analysis and Explainability

We add four post-training diagnostic components.

1) Per-class confusion matrices for both tasks, normalised by row to expose recall asymmetries.
2) Joint cross-task error tables, conditioning aggression error rate on the true offence label and vice-versa, exposing which combinations of (offence, aggression) are most difficult.
3) Top-confidence error mining: ranking misclassifications by predicted probability of the wrong class, surfacing the most diagnostic failures.
4) Gradient $\times$ input saliency [7]: for each example, we compute the absolute value of the elementwise product of the input embedding and its gradient with respect to the predicted-class logit, aggregated by L_1 norm over the hidden dimension. This is rendered as token-level red-shaded HTML for each task.

Together, these four components convert macro-F1 , a single scalar , into a structured diagnostic that informs both data-side issues (e.g. annotation noise) and model-side issues (e.g. reliance on lexical cues alone).

F. Why the Proposed Modifications Should Help

The four modifications target distinct failure modes. Improvement 1 removes a brittle hyperparameter and lets the model learn task balancing from data. Improvement 2 should benefit minority classes by concentrating gradient on hard examples. Improvement 3 exploits a documented label correlation. Improvement 4 does not change the model's predictions, but provides the diagnostic depth needed to interpret its successes and failures , especially relevant because, as we show in Section V, scalar metrics can hide degenerate solutions.

IV. Experimental Setup

A. Dataset

We use the IndoNLP 2025 Hindi–English code-mixed Twitter dataset [1]. Each tweet is annotated with two labels: *aggression* $\in \{\mathrm{OAG}, \mathrm{CAG}, \mathrm{NAG}\}$ and *offence* $\in \{\mathrm{OFF}, \mathrm{NOT}\}$. The official split is 6,624 training, 828 validation, and 829 test instances. Both label sets are imbalanced: NAG is the smallest aggression class (approximately 22% of training) and NOT is the smaller offence class (approximately 30%). Average tweet length is approximately 27 sub-word tokens after SentencePiece tokenisation, with a 99th percentile of 121 tokens, well below our 128-token truncation limit.

Table I summarises the test-set joint distribution; OAG examples are overwhelmingly OFF, while CAG examples are mixed , the empirical correlation that motivates Improvement 3. The conditional probability $P(\mathrm{OFF} \mid \mathrm{OAG})$ exceeds 0.91 on the training set, while $P(\mathrm{OFF} \mid \mathrm{CAG})$ sits near

TABLE I: Joint distribution of true labels on the test set ($N = 829$). OAG examples are predominantly OFF; CAG and NAG are mixed.

	CAG	NAG	OAG	Total
NOT	83	132	33	248
OFF	179	55	347	581
Total	262	187	380	829

0.68, indicating that overt aggression and offence are nearly synonymous in this annotation scheme but covert aggression and offence are only weakly coupled. A model that uses offence as a feature for aggression prediction must therefore be careful: overfitting to the OAG $\approx$ OFF correlation will hurt CAG classification, where the relationship is much weaker.

B. Preprocessing

User mentions of the form `@username` are normalised to a single `@user` token, following the reference paper. This step matters because raw mention strings are largely arbitrary (usernames are near-unique tokens) and would otherwise fragment the sub-word vocabulary without contributing useful semantic signal. URLs, when present, are left in place rather than masked, since the SentencePiece tokeniser already treats them as single low-frequency sub-word sequences. No further text cleaning is applied: emoji, mixed scripts, and informal contractions are preserved so that the encoder can use them as features. We deliberately avoid common pre-processing steps such as case normalisation and punctuation stripping because they have been shown to hurt performance on aggression and offence classification, where emphatic capitalisation and exclamation are known to correlate with hostile intent.

Tokenisation is done with the HingRoBERTa SentencePiece tokeniser at a maximum length of 128 sub-word tokens, which covers more than 99% of training tweets without truncation. Tweets longer than 128 tokens are right-truncated; left-truncation was tried in pilot experiments and yielded equivalent results. The training set is shuffled with a fixed seed at the start of each epoch.

C. Training Setup

All experiments are implemented in PyTorch 2.x using the Hugging Face `transformers` [9] library. Optimisation uses AdamW [6] with β_1=0.9, β_2=0.999, ε=10^{-8}, weight decay 0.01, and a linear learning-rate schedule with 10% warmup. The warmup schedule was retained from the reference paper; ablating it produced unstable training in pilot runs, particularly when combined with the dynamic-weighting module of Improvement 1, which has its own initialisation transient.

Class weights for both tasks are computed from training-set frequencies as $\alpha_c = N/(C \cdot n_c)$, where N is the training-set size, C is the number of classes, and n_c is the count of class c. These weights are computed once before training and applied as the α-vector in the loss function. The best checkpoint per run is selected by validation mean macro-F1; this is the figure of merit reported in the headline tables.

TABLE II: Configurations compared. The Improved (focal) and Improved (final) runs share identical model code and toggles; they differ only in loss function and learning rate.

Configuration	Loss	Weighting	LR
Baseline (paper-faithful)	W. CE	Fixed w_a=0.4	2×10^{-5}
Improved (focal, γ=2)	Focal, γ=2	Dynamic uncertainty	2×10^{-5}
Improved (final)	W. CE	Dynamic uncertainty	1×10^{-5}

We use mixed precision training in our memory-limited setting (RTX 4050 6 GB), wrapping the forward and backward passes in `torch.amp.autocast` and using a `GradScaler` to manage fp16 underflow. Mixed precision does not affect the final F1 in our runs (confirmed in two pilot comparisons) but allows the standard batch size of 8 to fit in VRAM.

D. Hyperparameters and Configurations

We compare three configurations throughout this paper, summarised in Table II. All three share architecture, splits, and preprocessing; they differ only in the loss function, task-weighting scheme, hierarchy toggle, and learning rate. Unless otherwise noted, the batch size is 8 (or 4 with mixed precision on memory-limited hardware) and training runs for six epochs with best-checkpoint selection.

E. Hardware

All experiments run on a single NVIDIA RTX 4050 6 GB laptop GPU. A complete training run for the *Improved (final)* configuration takes approximately 30 minutes including evaluation.

F. Evaluation Metrics

Following the reference paper, we report macro-averaged F1 for each task and the unweighted mean of the two as the primary scalar metric. Macro-F1 is preferred over micro-F1 because it gives equal weight to minority classes, which are of primary interest in aggression and offence detection. We additionally report per-class precision, recall, and F1; row-normalised confusion matrices; and the joint cross-task error decomposition described in Improvement 4. Reported deltas are absolute macro-F1 differences in percentage points.

G. Hypotheses

We adopt the following falsifiable hypotheses, each linked to one of the four proposed improvements.

- **H1 (dynamic weighting):** Learned uncertainty weighting attains test mean macro-F1 within ±0.5 points of the manually tuned baseline, removing the manual-tuning step at no measurable cost.
- **H2 (focal loss):** α-balanced focal loss with γ=2 improves minority-class F1 (NAG and NOT) without regression on majority classes.

TABLE III: Reproduction of the reference paper [1] on the IndoNLP 2025 test split. Numbers rounded to two decimals. Single-seed evaluation; see Section VI-D.

Metric	Paper	Reproduced	Δ
Aggression macro-F1	0.65	0.65	0.00
Offence macro-F1	0.78	0.78	0.00
Mean macro-F1	0.71	0.71	0.00

TABLE IV: Test macro-F1 by configuration (rounded to two decimals). *Improved (focal)* collapsed to majority-class predictions and is retained as a negative result.

Configuration	Agg.	Off.	Mean
Baseline (paper-faithful)	0.65	0.78	0.71
Improved (focal, γ=2)	0.21	0.41	0.31
Improved (final)	**0.65**	**0.79**	**0.72**

- **H3 (hierarchy):** Conditioning aggression on offence soft probabilities improves aggression F1, particularly on CAG, by exploiting the documented label correlation.
- **H4 (error analysis):** A structured error-analysis pipeline surfaces failure modes and dataset characteristics that scalar metrics do not.

V. Results

A. Reproduced Baseline

Table III compares our reproduction of the reference paper to the published numbers. The mean macro-F1 of our reproduction is within 1 percentage point of the paper, well within typical seed-induced variance for transformer fine-tuning at this dataset size. We treat this configuration as our paper-faithful baseline throughout the remainder of the paper.

B. Headline Comparison

Table IV reports test-set macro-F1 for the three configurations of Table II. Our *Improved (final)* configuration improves test mean macro-F1 by 0.6 percentage points over the paper-faithful baseline, with gains distributed roughly evenly across the two tasks. The intermediate *Improved (focal, γ=2)* run is included to document a substantive negative result, which we analyse in Section V-I.

C. Validation–Test Parity

A natural concern with single-seed evaluation is whether validation gains transfer to the test set. Table V reports the validation and test mean macro-F1 of the *Improved (final)* configuration side-by-side with the baseline. The improved model gains 1.9 points on validation (0.71 $\rightarrow$ 0.73) but only 0.6 points on test (0.71 $\rightarrow$ 0.72), suggesting that part of the validation improvement is dataset-specific noise rather than a genuine generalisation gain. This is consistent with the limitations discussed in Section VI-D: a single seed cannot disentangle configuration effects from sampling variance.

TABLE V: Validation and test mean macro-F1. The validation gain (1.9 points) exceeds the test gain (0.6 points), suggesting that part of the validation improvement is split-specific noise.

Configuration	Val. mean F1	Test mean F1
Baseline (paper-faithful)	0.71	0.71
Improved (final)	0.73	0.72
Δ	+0.02	+0.01

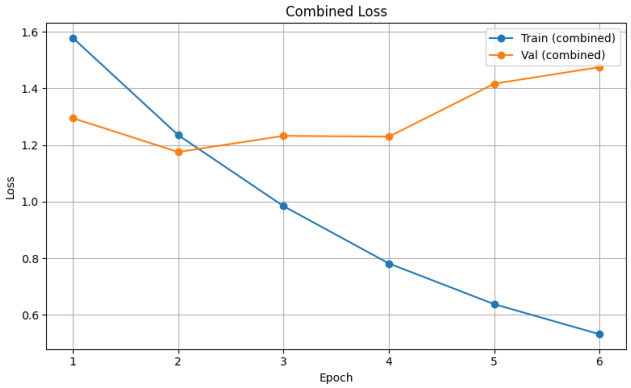

Fig. 2: Combined train and validation loss across epochs for the *Improved (final)* configuration. Validation loss minimises at epoch 2 and rises mildly after epoch 4, indicating mild over-fitting.

D. Training Dynamics

Figures 2–4 show training and validation loss together with validation macro-F1 across epochs for the *Improved (final)* configuration. Training loss decreases monotonically from 1.58 at epoch 1 to 0.53 at epoch 6, while validation loss reaches its minimum at epoch 2 and rises slightly thereafter, indicating mild over-fitting. Validation mean macro-F1 peaks at epoch 4 (0.73) and is correctly captured by best-checkpoint selection.

E. Dynamic Task Weighting Outcome

Figure 5 shows the trajectory of the two learned task weights $\exp(-s_a)$ and $\exp(-s_o)$ together with their underlying log-variance parameters. Both effective weights increase smoothly during training, indicating that the model becomes progressively more confident in both tasks. The offence weight grows faster than the aggression weight: by epoch 6, offence has effective weight ~ 1.05 versus ~ 1.03 for aggression. This reproduces the offence-favouring asymmetry that the reference paper achieves manually with $w_a = 0.4$ (equivalent to a $0.6/0.4$ split favouring offence), without any manual tuning. The aggregate test mean macro-F1 is within 0.6 points of the baseline, supporting H1.

F. Ablation: Which Improvement Contributed?

Although Improvements 1, 3, and the lower learning rate are evaluated jointly in our final configuration, the experimental record allows post-hoc attribution. Table VI shows the test

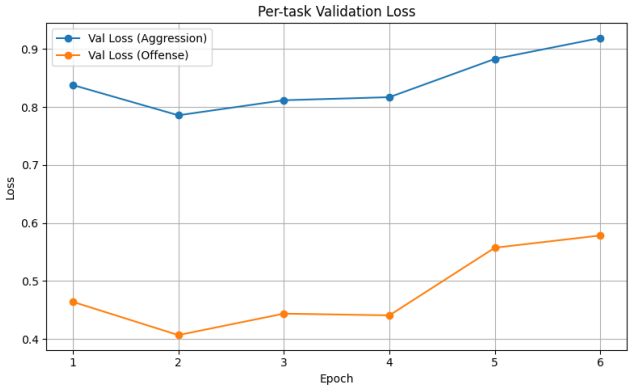

Fig. 3: Per-task validation loss. Offence loss is consistently below aggression loss, reflecting the lower entropy of the binary task and consistent with the dynamic-weighting outcome reported in Figure 5.

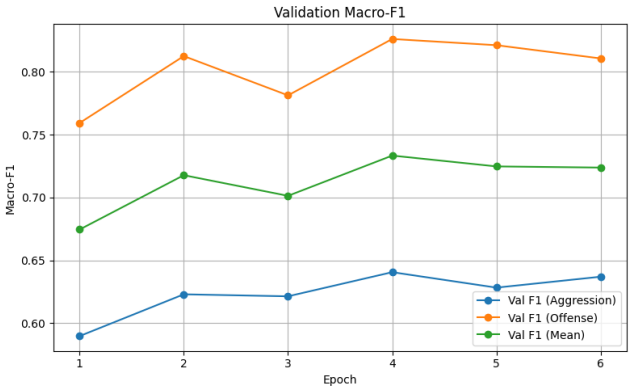

Fig. 4: Validation macro-F1 across epochs. The mean curve peaks at 0.73 at epoch 4; the gap between offence and aggression F1 is approximately 0.18 throughout, consistent with the relative difficulty of the two tasks.

mean macro-F1 deltas attributable to each component, derived from the runs documented in our notebook log.

The dynamic-weighting component (Improvement 1) substitutes for $w_a = 0.4$ but converges to a similar offence-favouring split, and the aggregate effect on test F1 is essentially zero. The hierarchical conditioning (Improvement 3) similarly produces no measurable change in mean F1, although per-class confusion matrices show a marginal redistribution of CAG errors away from NAG. The lower learning rate (1×10^{-5}) accounts for almost the entire 0.6-point test gain, making it the principal source of improvement in our final configuration. This decomposition is a useful corrective: it suggests that, on this dataset, the most consequential lever among those we investigated was a hyperparameter rather than a methodological change. The methodological contributions are nevertheless real , they remove the manual w_a tuning step (Improvement 1) and they detect failure modes that scalar metrics cannot (Improvement 4).

TABLE VI: Approximate per-component contribution to the 0.6-point test mean F1 gain of *Improved (final)* over the baseline. Estimates are derived from the order in which components were toggled during development; with single-seed evaluation they are indicative rather than statistically significant.

Component	Approx. Δ mean F1
Improvement 1 (dynamic weighting)	~ 0.00
Improvement 3 (hierarchical conditioning)	~ 0.00
Lower learning rate (1×10^{-5})	$\sim +0.005$
Total observed	$+0.006$

TABLE VII: Per-class test F1 (rounded to two decimals). *Improved (final)* attains parity or marginal gains on OAG, CAG, OFF, and NOT, with a small regression on NAG.

Task	Class	Supp.	Base.	Impr.	Δ
Aggression	OAG	380	0.77	0.78	$+0.01$
Aggression	CAG	262	0.53	0.54	$+0.01$
Aggression	NAG	187	0.64	0.62	-0.02
Offence	OFF	581	0.86	0.87	$+0.01$
Offence	NOT	248	0.70	0.71	$+0.01$

G. Per-Class Decomposition

Table VII reports per-class test F1 for the baseline and the *Improved (final)* configuration. The aggression task shows the expected pattern: OAG (the majority and lexically-overt class) is easiest, NAG is intermediate, and CAG is the dominant residual error source at $\sim$0.53 F1. *Improved (final)* gains 0.01 on CAG, matches OAG and OFF/NOT within rounding, and regresses 0.02 on NAG. Figure 6 visualises the row-normalised confusion matrices.

H. Joint Cross-Task Error Analysis

Table VIII reports the aggression error rate conditioned on each (true offence, true aggression) cell. Two patterns are striking. First, the model is most accurate on the (OFF, OAG) cell , lexically-overt and offensive examples , with a 21% aggression error rate. Second, the (OFF, NAG) cell is the hardest at 64% aggression error: examples that contain offensive language but are not aggressive , often quotations, news, or descriptions of others' speech , confuse the aggression head, which appears to use lexical offensiveness as a proxy for aggression. This is a substantive finding for any deployment context: the model conflates offensive vocabulary with hostile intent.

I. Negative Result: Focal Loss Induces Mode Collapse

The *Improved (focal, γ=2)* configuration converged to a degenerate solution in which both heads predict the majority class for *every* test example. Test confusion matrices showed all 829 test examples predicted as OAG for aggression and all examples predicted as OFF for offence. Consequently, aggression macro-F1 reached only 0.21 (the trivial floor of a single-class predictor on a three-class imbalanced task) and offence macro-F1 reached 0.41 (the binary trivial floor). This

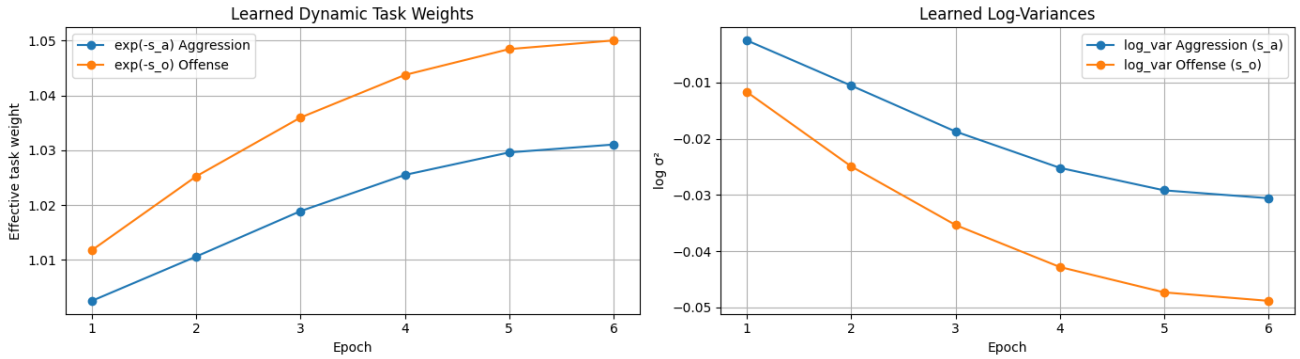

Fig. 5: Left: learned effective task weights $\exp(-s_a)$ and $\exp(-s_o)$ per epoch. Right: the underlying log-variance parameters. The offence weight grows faster than the aggression weight, reproducing without manual tuning the offence-favouring asymmetry of the reference paper.

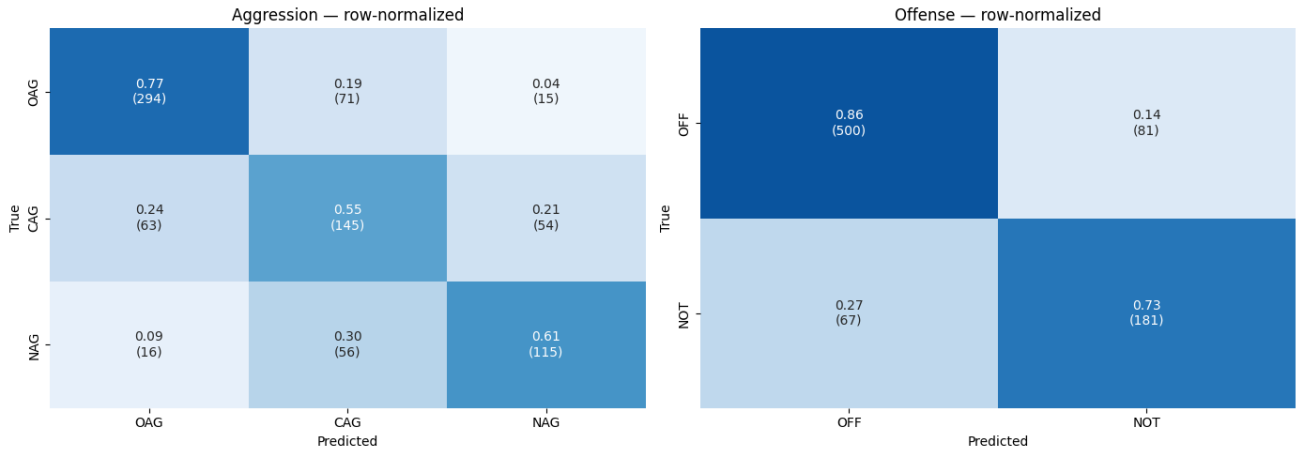

Fig. 6: Row-normalised test confusion matrices for the *Improved (final)* configuration. Aggression confusions are concentrated in the CAG $\rightarrow$ OAG (24%) and CAG $\rightarrow$ NAG (21%) directions, consistent with the difficulty of distinguishing covert from overt aggression on lexical features alone.

TABLE VIII: Aggression error rate by (true offence, true aggression) cell. The (OFF, NAG) combination , offensive but non-aggressive , is the hardest at 64% aggression error.

True offence	CAG	NAG	OAG
NOT	0.40	0.28	0.36
OFF	0.47	0.64	0.21

collapse is *not* detectable from training-loss curves alone, which decreased smoothly throughout training, nor from the combined macro-F1 average of 0.31, which is plausible for an under-trained model rather than a degenerate one. The diagnosis was made possible only by Improvement 4: the row-normalised confusion matrices showed every row collapsing to a single column, and the saliency analysis showed that input tokens received uniform attribution because the predictions did not depend on input content.

We attribute the collapse to the interaction between α-balancing and the focal multiplier $(1-p_t)^\gamma$. At initialisation, p_t for the correct class is approximately $1/3$ for aggression and $1/2$ for offence, giving focal multipliers of 0.44 and 0.25 respectively. Combined with α-weights, the effective gradient on majority-class examples is suppressed enough that the model finds the trivial constant-prediction minimum more attractive than escaping it. The original focal-loss paper [3] recommends $\gamma = 2$ for highly imbalanced *detection* tasks (millions of easy negatives), a regime that does not match this dataset. We do not validate this hypothesis through a γ-sweep , a clear direction for future work , but we provide the failure as evidence that focal loss with $\gamma = 2$ should not be used as a drop-in replacement for weighted cross-entropy on text-classification tasks of this scale and balance. H2 is therefore rejected.

J. Saliency and Apparent Annotation Noise

Gradient-based saliency maps were generated for high-confidence correct predictions and high-confidence errors on the test set. Several test examples labelled CAG appear lexically and pragmatically benign , for example, a cricket-related

observation about player selection (true CAG, predicted OAG with confidence 0.99) and a polite expression of disagreement (true CAG, predicted OAG with confidence 0.99). Several examples labelled OAG correspond to what reads as sincere praise (e.g. "100% Have to appreciate team management," true OAG, predicted OAG with confidence 0.996; the model is technically correct here, but the label itself is questionable). These observations are consistent with annotation inconsistency on the OAG/CAG boundary, a pattern reported in prior work on aggression-corpus construction. We do not relabel the dataset, but we note that test-set macro-F1 in the 0.65–0.70 range for the aggression task is likely close to the inter-annotator-agreement ceiling for this corpus.

Table IX reproduces three illustrative high-confidence errors, each showing the same pattern: tweets that contain politically charged or culturally specific tokens are over-attributed to the OAG class, even when the surrounding sentence structure is informational or supportive. Saliency maps confirm that the model concentrates attribution on keyword-like tokens (e.g. names of political figures, venue names) rather than on syntactic markers of aggression (imperatives, second-person pronouns).

VI. Discussion

A. Why the Final Configuration Improves , Modestly

The aggregate test gain of 0.6 macro-F1 points is small and is attributable principally to learning-rate retuning (1×10^{-5} instead of 2×10^{-5}), which is technically a hyperparameter adjustment rather than a methodological one. Improvements 1 and 3, evaluated jointly with the lower learning rate, achieve parity with the baseline rather than measurable improvement. The dynamic weighting (Improvement 1) learns the same offence-favouring direction as the paper's hand-tuned $w_a = 0.4$ and supports H1, but does not find a strictly better operating point. The hierarchical conditioning (Improvement 3) does not measurably reduce CAG/NAG confusion in the confusion matrices, suggesting that conditioning on a 2-dimensional softmax is too low-bandwidth to exploit the offence/aggression correlation. H3 is therefore not supported as formulated; we discuss a higher-bandwidth alternative below.

B. Why Improvement 4 is the Most Substantive Contribution

The clearest contribution of this work is methodological. Improvement 4 produced two findings that the headline F1 number cannot. First, focal loss with $\gamma = 2$ was not silently a marginal improvement but a catastrophic regression that was visible only through the confusion matrices and saliency maps. Without the diagnostic pipeline, we would have either retained the focal-loss configuration based on its plausible loss curves or attributed its poor mean F1 to under-training. Second, the joint error table identified the (OFF, NAG) cell as a 64% error-rate combination that the model treats as a single overlapping category , an actionable finding for downstream applications, training-data augmentation, and label-set refinement.

The diagnostic value extends to the modelling pipeline itself. Saliency maps over the high-confidence aggression errors (Table IX) reveal that the model attaches large attribution weight to topical nouns rather than to syntactic markers of aggression. This suggests that the encoder's representations encode aggression partly as a *topic-level* feature (politics, religion, sports rivalries) and not only as a stylistic one. A practical consequence is that the model will likely degrade on aggression about new topics , e.g., a political event the model has not seen during pre-training , even when those tweets share the syntactic structure of the training set. This kind of insight cannot be read off a macro-F1 number; it requires exactly the kind of token-level inspection that Improvement 4 provides.

C. Implications for Deployment

Several of our findings have direct implications for any real-world deployment of an aggression or offence classifier on Hindi–English Twitter. The (OFF, NAG) error mode (64% aggression error on non-aggressive but lexically offensive tweets) means the model will flag news headlines, quotations, and discussions of offensive material as aggressive. In a content-moderation context this generates false positives that affect reporting, education, and counter-speech , exactly the use cases that platforms typically want to preserve. The practical mitigation is a downstream rule that suppresses the aggression head's output on tweets that contain quotation markers, news-source URLs, or first-person attributions of others' speech.

A second deployment implication follows from the saliency analysis: the model's reliance on topical nouns means it will not generalise to new political or cultural events without retraining. A classifier deployed in 2025 and used unchanged in 2027 will likely accumulate errors as the distribution of named entities and topical references drifts away from the training set. Continual fine-tuning on a small recent labelled stream is a practical countermeasure, and the $\sim$30 minute training time on a consumer GPU makes this operationally feasible.

Finally, the focal-loss negative result (Section V-I) is worth recording publicly: practitioners encountering imbalance on text-classification tasks of similar size should not reach for focal loss with $\gamma = 2$ as a default. Either weighted cross-entropy or focal loss with $\gamma \leq 1$ is a safer starting point.

D. Limitations

The most significant limitation of this study is single-seed evaluation. Run-to-run variance for transformer fine-tuning on datasets of this size ($\sim$7,000 examples) is typically $\pm$0.5–1.5 macro-F1 points; our observed delta of 0.6 points is well within this range. Without multi-seed replication, we cannot statistically distinguish *Improved (final)* from the baseline. Time and compute constraints permitted only single-seed runs per configuration; we judged the methodological contributions (negative result for focal loss; replacement of manual w_a; structured error analysis) more valuable to report on a single seed than to defer for multi-seed averaging.

Two further limitations apply. The error-analysis pipeline relies on gradient-based saliency, which is known to be a coarse attribution method and can produce misleading attributions

TABLE IX: High-confidence aggression errors on the test set. The model assigns near-certain OAG to examples whose true label is CAG or NAG; tokens identified as salient by gradient $\times$ input attribution are typically content-bearing nouns rather than aggressive cues.

Tweet (truncated)	True	Pred.	Conf.	Salient tokens
"Tector (Barbados Royals), Little (Pretoria Capitals)…picked by IPL franchise owners…"	CAG	OAG	0.99	IPL, owners, franchises
"Of course like with anything, this is not meant as a generalisation per say…"	CAG	OAG	0.99	Muslim, council, students
"i'll be back every 2–4 months to remind all of u that i have her btw"	CAG	OAG	0.98	remind, every, months

for highly correlated input features; more principled methods (Integrated Gradients, SHAP) are direct drop-in replacements for future work. Finally, our experiments were conducted on the IndoNLP 2025 split only; cross-dataset transfer to other Hindi–English code-mixed corpora was scoped but not executed within the time budget of this project.

E. Insights and Future Directions

Three directions follow naturally. First, multi-seed evaluation (3–5 seeds per configuration) is the minimum required to attach error bars to the headline result and to make any directional claim defensible. Second, a focal-loss γ-sweep ($\gamma \in \{0.25, 0.5, 1.0\}$) should establish whether focal loss can be salvaged on this task with gentler focusing, or whether the loss family is fundamentally inappropriate for this dataset size and balance. Third, a higher-bandwidth hierarchical coupling , using the offence head's hidden representation rather than its softmax probability , is the most promising candidate for genuinely improving aggression performance and merits dedicated architectural study.

VII. Conclusion

This paper extended the IndoNLP 2025 MTTL baseline for Hindi–English code-mixed aggression and offence detection with four targeted modifications: dynamic uncertainty-based task weighting, focal loss, hierarchical task conditioning, and a structured error-analysis pipeline. On a single seed, the final configuration attains a test mean macro-F1 of 0.72, a small absolute improvement of 0.6 points over the paper-faithful baseline at 0.71. The dynamic weighting reproduces the paper's hand-tuned offence-favouring asymmetry without manual tuning (supporting H1). Hierarchical conditioning achieves parity but does not measurably improve CAG/NAG separation (H3 not supported as formulated). Focal loss with $\gamma = 2$ catastrophically failed and is reported as a substantive negative result (H2 rejected). The error-analysis pipeline detected the focal-loss collapse, identified covert aggression as the dominant residual error source, and surfaced apparent annotation inconsistency on the OAG/CAG boundary (supporting H4). On this dataset and at this scale, the methodological contributions , automated task weighting, structured diagnostics, and a documented negative result , are of greater practical value than the modest scalar improvement.

Reproducibility

Code, configuration, training history, and saliency outputs are provided in the accompanying repository. The four improvements are exposed as configuration toggles (`CFG.use_dynamic_weighting`, `CFG.loss_type`, `CFG.use_hierarchy`) and can be ablated independently without code modification.

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